

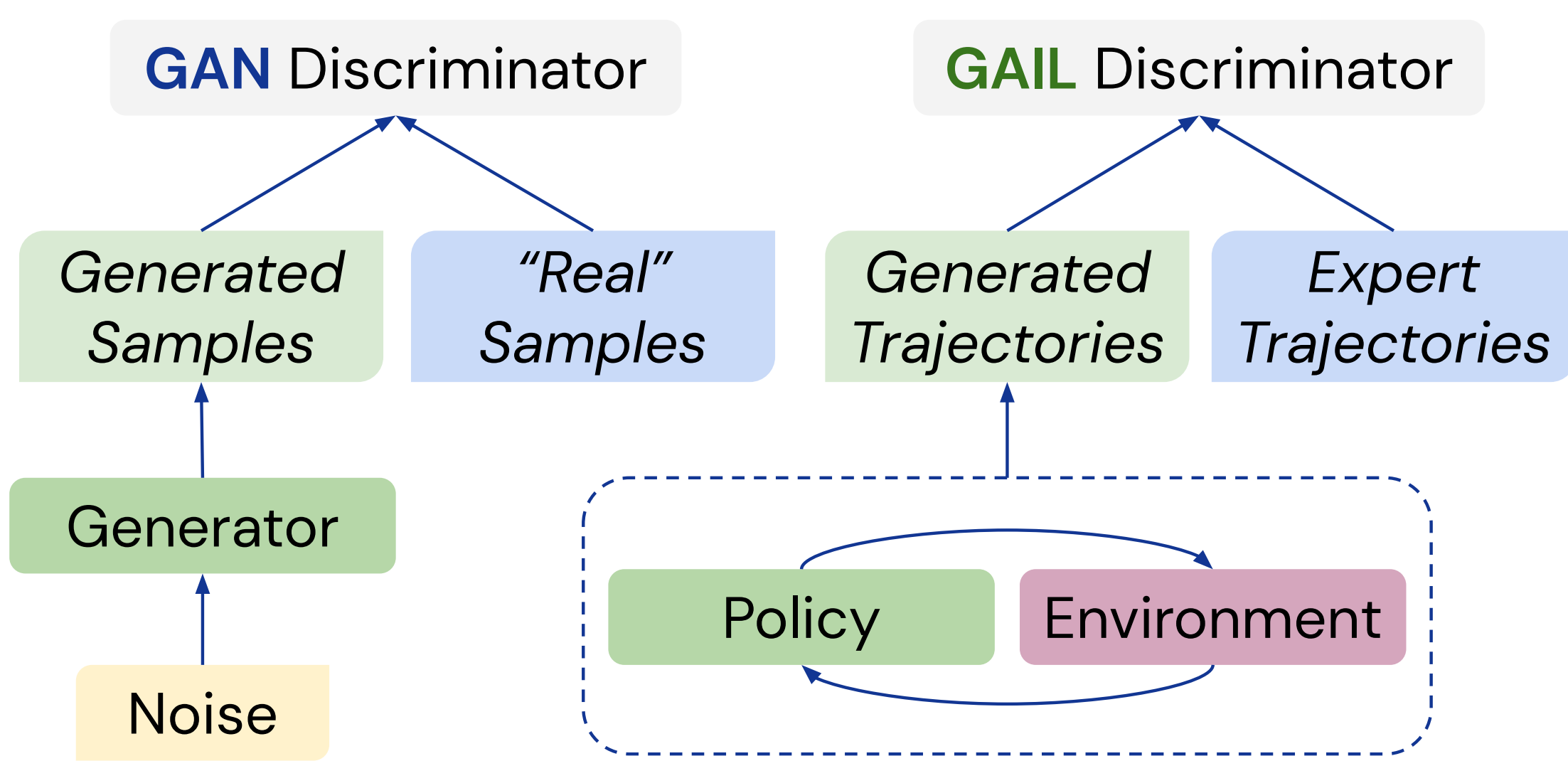
COMBATING FALSE NEGATIVES IN ADVERSARIAL IMITATION LEARNING

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Summary

In **Generative Adversarial Imitation Learning (GAIL)** [Ho and Ermon, 2016], inspired by GANs, a discriminator is trained to differentiate agent episodes from expert demonstrations representing the desired behavior, and is then used as a reward function.



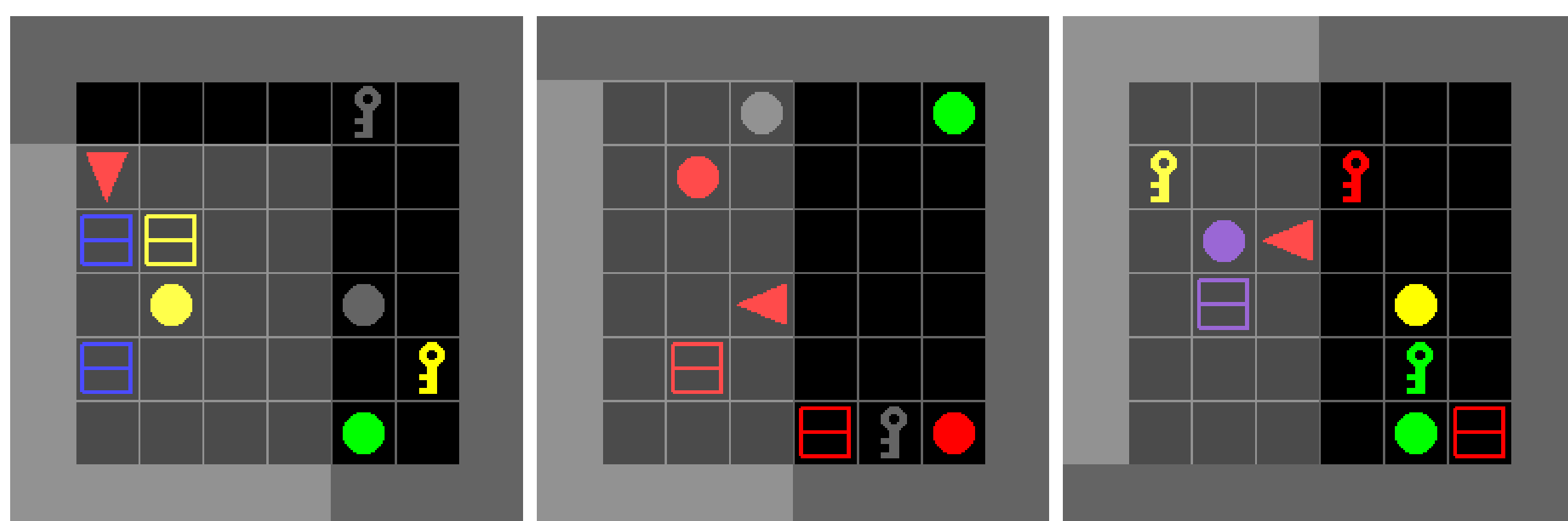
As the trained policy learns to be more successful, **the agent trajectories** (negative examples for discriminator) **become increasingly similar to expert ones**.

We show that these **False Negatives** (i.e. successful agent episodes) impede discriminator's learning and significantly hinder adversarial imitation learning.

We propose a method, **Fake Conditioning**, to alleviate the impact of false negatives and test it on the BabyAI environment.

Baby AI Environment

We use the **BabyAI platform** [Chevalier-Boisvert et al., 2018] as the testbed since it is well-suited to analyze imitation learning's sample efficiency.



GoToLocal
go to the yellow key

PickupLoc
*pick up a box
in front of you*

PutNextLocal
*put the green ball
next to the red key*

Conditioned Recurrent Discriminator

Memory

For partially-observable environments it is necessary to equip **the discriminator with a memory** to model the true environment reward. To this end, we implement the discriminator as **a recurrent neural network**.

Conditioned Discriminator

To address the goal-conditioned nature of the problem, **the discriminator is conditioned on the instruction c along with the trajectory τ** . The resulting training objective is

$$\max_{\pi} \min_D \left(\mathbb{E}_{\pi}[-\log(1 - D(c, \tau))] + \mathbb{E}_{\pi_E}[-\log D(c, \tau)] + \lambda H(\pi) \right).$$

We use two finite buffers B_{agent} and B_{expert} that contain agent trajectories and expert demonstrations, respectively, to compute a batched version of **the discriminator loss**:

$$L_D(\theta) = \mathbb{E}_{(c, \tau) \sim B_{agent}} -\log(1 - D_{\theta}(c, \tau)) + \mathbb{E}_{(c, \tau) \sim B_{expert}} -\log(D_{\theta}(c, \tau)).$$

False Negatives

While GAIL works well in the initial phase of the learning procedure, **as the agent reaches high success rates, its performance tends to be unstable**.

The problem is due to the fact that **the discriminator has to classify successful agent episodes as negative examples**, leading to incorrect rewards.

We refer to this phenomenon as the **False Negatives (FN)** problem, as successful agent trajectories are falsely labeled as negative examples.

Oracle Filtering

Oracle Filtering (OF) is our diagnostic method to measure FN problem's strength.

OF uses the environment's true reward signal to **identify the successful agent trajectories and filter them out**.

Fake Conditioning

By replacing the original instruction with a random one, we can produce a new instruction-trajectory pair that is unsuccessful. We call this technique **Fake Conditioning (FC)**.

We apply FC to generate **an alternative set of true negatives** which are always unsuccessful, and then use them to train the discriminator. The new set can be based on fake conditioned expert demonstrations (**Expert FC**), or agent episodes (**Agent FC**).

Auxiliary Rewards

We train **a second discriminator** to penalize the agent for exploiting potential inaccuracies of the first one. We introduce two variants.

Done Detector is trained to predict if a given expert trajectory is complete or not. **Blank Conditioning** is input no instructions to focus on general behaviour patterns.

Positives and negatives used to train different variants of a discriminator.

	Conventional	Agent FC	Expert FC	Done Detector	Blank Conditioning
P trajectories	expert				expert
<i>instructions</i>	<i>original</i>				<i>blank</i>
N trajectories	agent	agent	expert	incomplete expert	agent
<i>instructions</i>	<i>original</i>	<i>fake</i>	<i>fake</i>	<i>original</i>	<i>blank</i>

Results

We report **success rate** and consider a given task to be **solved if 99%** is achieved.

Three expert demonstrations sets of different sizes are used. The largest set is the minimum necessary demonstration set needed to solve the task using Behavioral Cloning (BC). The other two are **8** times and **64** times smaller.

Model	GoToLocal			PickupLoc			PutNextLocal		
	$\frac{1}{64}$	$\frac{1}{8}$	1	$\frac{1}{64}$	$\frac{1}{8}$	1	$\frac{1}{64}$	$\frac{1}{8}$	1
Behavioral Cloning	67.3	90.8	99.9	65.2	95.9	100.0	26.8	77.9	99.9
Baseline GAIL	52.3	86.6	98.7	60.4	90.4	98.2	26.9	80.8	94.9
Oracle Filtering	99.5	99.5	99.7	98.2	99.5	99.5	66.7	93.1	97.4
Our method	98.9	99.7	99.8	96.5	99.2	99.3	98.2	99.6	99.3

Using **OF significantly improves the performance** which confirms that false negatives have a detrimental effect on GAIL training.

Our method solves all tasks with 8 times fewer demonstrations than BC.

Acknowledgments

Konrad Żołna is supported by the National Science Center, Poland (2017/27/N/ST6/00828, 2018/28/T/ST6/00211). This research was mostly performed at Mila with funding by the Government of Quebec and CIFAR, and enabled by Compute Canada (www.computeCanada.ca).